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Professor Michael Böhm
Editor-in-Chief
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Dear Professor Böhm,

We are pleased to submit our manuscript, "***Shared Genetic Architecture of Body Mass Index and Heart Failure at the FTO Locus***", for consideration in the *European Journal of Heart Failure*. We show that, at the *FTO* locus, the heart-failure association is shared with the primary body mass index (BMI) component and points to a haplotype overlapping adipogenic regulatory DNA.

Obesity is a major modifiable contributor to heart failure, and incretin-based therapies now benefit patients with obesity-related heart failure with preserved ejection fraction. Mendelian randomisation supports a causal role for BMI, but genome-wide estimates do not reveal which loci or biological processes carry this effect. At the *FTO* locus (16q12.2), associations with both BMI and heart failure have been reported, yet it remains unclear which BMI-associated component shares the heart-failure signal and how it relates to adipogenic regulatory biology.

We therefore integrated genome-wide data for BMI (GIANT) and heart failure (HERMES) with external regional evaluation in FinnGen, complementary observational analyses in UK Biobank and adipogenic chromatin annotation. The principal findings are:

1. **Systematic prioritisation.** We grouped 78 BMI instruments into 75 genomic regions; five of these showed BMI–heart-failure colocalisation in both HERMES releases. Of the five, the heart-failure effect at 16q12.2 was closest to that predicted from its BMI effect under the genome-wide BMI–heart-failure relationship.
2. **Component-level resolution.** The primary BMI component at 16q12.2 colocalised with the heart-failure signal at this locus (posterior probability 0.974), and regional analysis in FinnGen provided external support for a shared signal (posterior probability 0.907).
3. **Adipogenic regulatory context.** Fine-mapping and chromatin annotation prioritised the rs1558902–rs1421085 haplotype, which overlaps regulatory DNA active during adipogenesis; rs1421085 itself has been shown experimentally to alter adipocyte thermogenesis.

Together, these analyses move the *FTO*–heart-failure association from a shared regional signal to a specific BMI component and its adipogenic regulatory context. As obesity becomes a therapeutic target in heart failure, this locus-level evidence is directly relevant to EJHF readers.

This manuscript reports original work that has not been published in any other journal and is not under consideration elsewhere. All authors have read and approved the submitted manuscript. The authors declare no conflicts of interest. ChatGPT (OpenAI) and Claude (Anthropic) assisted with language editing and the preparation of submission materials; the authors reviewed and verified all output and take full responsibility for the scientific content and its interpretation.

Thank you for considering our manuscript.

Yours sincerely,

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